

Coding & Solution

Date	03 July 2026
Team ID	SWTID-2026-1614
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

- 1 Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
- 2 The solution should address the core problem statement and implement the key functional requirements.
- 3 Include all source files, configuration files, and setup instructions needed to run the application.
- 4 Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/v-harishkumar/hdi-predictor-ml-flask (or your custom repo URL)
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask (backend API), Jinja2 (HTML templating)
Key Features Implemented	• ML-based HDI Prediction: Input Life Expectancy, Schooling, and GNI to compute predicted HDI. • Development Tier Mapping: Automatically classifies predictions into developmental tiers (Low, Medium, High, Very High). • Country Search Autocomplete: Search bar for 195 countries to auto-fill forms with historical baseline values. • Premium Glassmorphism UI: High-contrast, mobile-responsive client-side layout. • Cloud Deployment Ready: Configured for simple deployment on Python-Flask servers (Render, PythonAnywhere).
Pending / Incomplete Features	• Interactive Map: Geographic country selection interface. • Multi-Country Comparison: Grid to view and compare different simulation runs side-by-side. • User Authentication: Session profiles to save history. • Live UNDP API Integration: Real-time indicators syncing.
Setup / Run Instructions	1 Clone the repository and open the project directory. 2 Install dependencies: <code>pip install -r requirements.txt</code> 3 Ensure model files (hdi_model.pkl or similar) are placed in the Flask/ or model/ folder. 4 To run locally: <code>flask --app Flask/app.py run</code> (or <code>python Flask/app.py</code>). 5 To deploy: Set up a web service on Render pointing to the main Flask script.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The HDI Predictor successfully implements the core ML-based socio-economic prediction flow end-to-end and is functional on the web platform. The multi-country comparison grid and live UNDP API integrations are scheduled as future enhancements for upcoming version releases.